

From a Gravity-Calibrated Helix to a Conditional Fine-Structure Correspondence

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Abstract. We report a frozen numerical correspondence generated from the dimensionless helix parameter established in the preceding TEF preprint. That construction uses the measured Newton constant to calibrate a primitive radius, $R_G = \sqrt{hG/(8\pi c^3)} = \ell_P/2$, and then fixes the dimensionless shape by one-cycle phase closure, $q\sqrt{1+q^2} = 2/\pi$, giving $q = 0.5563241800\dots$ without electroweak input. The present paper introduces the scale-free helical path excess $\delta_h = \sqrt{1+q^2} - 1$ and explores a conditional spinorial completion of a framed state by $SU(2) \simeq S^3$. We explicitly separate this completion, the minimal Hopf-coordinate identification $\eta = \beta$ with $q = \tan \beta$, and the choice of a unit round S^3 with its unnormalized Riemann volume measure. Under these assumptions the Hopf-band volume fraction is exactly $\sin^2 \beta$, while the unit-sphere volume is $2\pi^2$. We then *postulate, not derive*, the electromagnetic correspondence $\alpha_{\text{TEF}} = \delta_h/(2\pi^2)$, obtaining $\alpha_{\text{TEF}}^{-1} = 136.7621663\dots$, about 0.2002% from the 2022 CODATA recommended low-energy value. The factor $1/(2\pi^2)$ is not claimed to follow physically from probability normalization: it depends on the metric scale and measure convention unless a future TEF action fixes them. The result is therefore presented as a frozen, gravity-anchored geometric correspondence with explicit theoretical failure conditions, rather than a derivation or precision prediction of QED. Its main purpose is to expose a concrete possible map $G \rightarrow R_G \rightarrow q \rightarrow \alpha_{\text{TEF}}$ and, equivalently after calibration, a simple expression of the fine-structure candidate in terms of the inherited q .

1 Introduction

The fine-structure constant is the dimensionless electromagnetic coupling. In SI notation,

$$\alpha = \frac{e_{\text{SI}}^2}{4\pi\epsilon_0\hbar c}. \quad (1)$$

The 2022 CODATA recommended low-energy value is [2, 3]

$$\alpha(0)^{-1} = 137.035999177(21). \quad (2)$$

This value is a recommended adjustment informed by several precision inputs, rather than a single direct experiment. Electron magnetic-moment measurements and atom-recoil determinations provide complementary high-precision routes to the same low-energy coupling.

Beyond the Thomson limit, the electromagnetic coupling runs through vacuum polarization and radiative corrections. Electroweak quantities conventionally described by a weak mixing angle are likewise scale- and scheme-dependent [4]. Importantly for the present paper, modern lattice-QCD calculations treat the hadronic running of the electromagnetic coupling and the electroweak mixing angle within one correlated framework. Cè *et al.* computed both $\Delta\alpha_{\text{had}}$ and the hadronic contribution to the running weak angle from vacuum polarization functions [7]; Conigli *et al.* recently extended this program to permille-level precision for spacelike virtualities up to the multi-GeV regime and a high-precision

continuation toward the Z pole [8]. These results reinforce a basic constraint on any TEF interpretation: α and θ_W cannot be assigned unrelated scale corrections. Consequently, a proposed geometric number can be compared directly with Eq. (2), but a claim that it is an ultraviolet or “bare” seed requires a specified matching scale, renormalization prescription, and common electroweak dynamics.

Attempts to explain α by geometric or algebraic constants have a long history. The main methodological danger is not arithmetic error but post-hoc functional choice: sufficiently many simple combinations can approximate a known number. The criticism of Wyler-type formulae remains relevant in this respect [15]. The present work therefore separates two facts that must not be conflated. First, the parameter q was already frozen in an earlier publicly released TEF ansatz and is not reoptimized here [1]. Second, the map from q to α proposed below was recognized after comparison with electromagnetic data and therefore carries a look-elsewhere burden of its own.

The preceding TEF paper, *A Helical Spacetime Ansatz Linking the Planck Scale and Electroweak Mixing*, version 5.1, develops a gravity-calibrated helical spacetime ansatz and a candidate primitive geometric reference angle for weak mixing [1]. The present paper asks a narrower continuation question: can the same frozen helical shape support a simple, explicitly conditional correspondence with the electromagnetic coupling?

This question is adjacent to, but distinct from, several mainstream research programs in which low-energy couplings may be controlled by deeper structure. Quantum-gravity fixed points can constrain gauge couplings in asymptotic-safety scenarios [10]. Recent functional-renormalization-group work continues to calculate gravitational contributions to gauge- and Yukawa-coupling beta functions and to quantify their gauge- and regulator-dependence [11]. Other work asks whether familiar gauge symmetries or spacetime itself may be emergent rather than fundamental [12–14]. These approaches do not validate TEF. They establish only that the broader question—whether measured gauge parameters can encode ultraviolet geometric or gravity-related structure—is a recognizable field-theoretic one.

2 Inherited Result I: the gravity-calibrated helix

The present paper inherits the three postulates and two geometric propositions of Ref. [1]. They are summarized here so that the input chain is self-contained. A local spatial line is represented by the right-handed helix

$$\mathbf{r}(\phi) = (R \cos \phi, R \sin \phi, a\phi), \quad q \equiv \frac{a}{R} = \tan \beta. \quad (3)$$

The earlier paper first calibrates the transverse scale through

$$G = \frac{8\pi c^3 R^2}{h}, \quad (4)$$

so that the measured Newton constant gives

$$R_G = \sqrt{\frac{hG}{8\pi c^3}} = \frac{\ell_P}{2}. \quad (5)$$

Equation (4) is a TEF calibration postulate, not a prediction of G .

A complete turn has axial extension and internal helical length

$$P = 2\pi q R, \quad (6)$$

$$L_{\text{cyc}} = 2\pi R \sqrt{1 + q^2}. \quad (7)$$

The earlier ansatz assigns the gravitational line-tension scale c^4/G to the net axial extension,

$$E_{\text{cyc}} = \frac{c^4}{G} P, \quad (8)$$

while the internal traversal time is

$$T_{\text{cyc}} = \frac{L_{\text{cyc}}}{c}. \quad (9)$$

Its third postulate requires one primitive cycle to close its

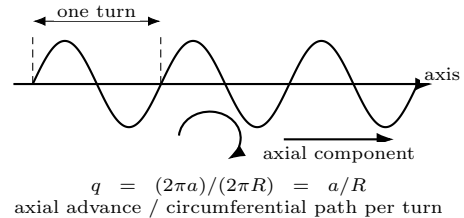


Figure 1: Schematic projection of the inherited TEF helix. The drawing is kinematic and not to scale. The same frozen shape parameter q is used throughout the present paper.

quantum phase,

$$\frac{E_{\text{cyc}} T_{\text{cyc}}}{\hbar} = 2\pi. \quad (10)$$

Before the gravity calibration is inserted, Eqs. (4)–(10) lead to

$$q\sqrt{1 + q^2} = X_G, \quad X_G \equiv \frac{hG}{4\pi^2 c^3 R^2}, \quad (11)$$

and therefore

$$q(G, h, c; R) = \sqrt{\frac{\sqrt{1 + 4X_G^2} - 1}{2}}. \quad (12)$$

This equation retains the explicit gravitational ancestry of the dimensionless shape. With $R = R_G$, however, the dimensional constants cancel and

$$q\sqrt{1 + q^2} = \frac{2}{\pi}, \quad (13)$$

so that

$$q^2 = \frac{-1 + \sqrt{1 + 16/\pi^2}}{2}, \quad q = 0.556324180045 \dots \quad (14)$$

The corresponding intrinsic angle is

$$\beta = \arctan q = 29.0882455^\circ, \quad (15)$$

$$\sin^2 \beta = 0.2363477652 \dots \quad (16)$$

No electromagnetic or weak-mixing datum is used in Eqs. (4)–(16). The important qualification is that G calibrates the microscopic scale, while the final calibrated shape is dimensionless and no longer depends numerically on the measured magnitude of G .

3 Proposition 1: helical path excess

For one turn, the circumferential projection has length $L_{\text{circ}} = 2\pi R$, while the helical path has length $L_{\text{helix}} = 2\pi R \sqrt{1 + q^2}$. Their scale-free fractional excess is

$$\delta_h \equiv \frac{L_{\text{helix}} - L_{\text{circ}}}{L_{\text{circ}}} = \sqrt{1 + q^2} - 1. \quad (17)$$

For the inherited value of q ,

$$\delta_h = 0.144332378858 \dots \quad (18)$$

Its small- q expansion begins at second order,

$$\delta_h = \frac{q^2}{2} - \frac{q^4}{8} + \mathcal{O}(q^6). \quad (19)$$

This makes δ_h a simple intensity-like scalar associated with the orthogonal helical component, but no QED amplitude or action coefficient is derived from Eq. (17).

4 Postulate S1: spinorial orientation completion

A circular helix embedded in ordinary three-space does not imply S^3 . The present paper makes an additional structural assumption: a framed TEF state retains a 4π orientation periodicity, so that its orientation variable is lifted from $SO(3)$ to the double cover $SU(2)$. As a manifold, the unit-quaternion representation of $SU(2)$ is S^3 [9].

This completion concerns the configuration/orientation variable of a framed state. It is distinct from the 2π quantum phase closure in Eq. (10): the latter closes one $U(1)$ phase cycle, whereas the former concerns a double-covered rotational configuration space. The present paper does not derive why TEF must retain the $SU(2)$ double-cover information; this is Postulate S1.

It is also important to distinguish three related but non-identical objects: a group element $U \in SU(2)$ describing orientation, a spinor transforming in an $SU(2)$ representation, and a projective pure state after an overall $U(1)$ phase has been quotiented out. The last object is $\mathbb{CP}^1 \simeq S^2$, not S^3 . Here S^3 is used specifically as the double-cover orientation/group manifold, not as a claim that every physical spin state lives on an observable three-sphere.

5 Postulate S2 and Proposition 2: Hopf identification

Using Hopf coordinates on the unit three-sphere,

$$z_1 = e^{i\xi_1} \cos \eta, \quad z_2 = e^{i\xi_2} \sin \eta, \quad (20)$$

with

$$0 \leq \eta \leq \frac{\pi}{2}, \quad 0 \leq \xi_1, \xi_2 < 2\pi, \quad (21)$$

the round unit- S^3 metric is

$$ds^2 = d\eta^2 + \cos^2 \eta d\xi_1^2 + \sin^2 \eta d\xi_2^2. \quad (22)$$

We now make a second structural assumption:

$$\boxed{\eta = \beta, \quad q = \tan \beta.} \quad (23)$$

This is the minimal parameter-preserving identification between the external helical ratio and the internal Hopf amplitude ratio. It introduces no new continuous parameter, but it is nevertheless a mapping choice that must ultimately be derived from the framed-state geometry. Hopf coordinates do not force Eq. (23), and the choice of the (z_1, z_2) decomposition presupposes a physical structure that fixes the relevant basis.

For the round metric, the Riemannian volume element is

$$dV = \sin \eta \cos \eta d\eta d\xi_1 d\xi_2, \quad (24)$$

and the total unit-sphere volume is

$$\text{Vol}(S^3) = 2\pi^2. \quad (25)$$

Conditional on Postulate S2, the volume fraction of the Hopf band $0 \leq \eta \leq \beta$ is

$$\frac{V(0 \leq \eta \leq \beta)}{\text{Vol}(S^3)} = \frac{4\pi^2 \int_0^\beta \sin \eta \cos \eta d\eta}{2\pi^2} \quad (26)$$

$$= \boxed{\sin^2 \beta}. \quad (27)$$

This is an exact mathematical identity under the identification $\eta = \beta$. It is best regarded as a conditional geometric representation of the weak-angle candidate, not as statistically independent evidence for it. In particular, for a point uniformly distributed on unit S^3 , the variable $|z_2|^2 = \sin^2 \eta$ is uniformly distributed on $[0, 1]$, so Eq. (27) is also the corresponding cumulative-volume identity.

6 Convention C1: metric and measure

The appearance of $2\pi^2$ requires a convention that must be separated from physics. Under Convention C1 we use the unit round metric of Eq. (22) and the corresponding *unnormalized* Riemannian volume measure dV . A constant scalar Laplace–Beltrami mode

$$\Psi_0(U) = C, \quad U \in S^3, \quad (28)$$

normalized with respect to this measure satisfies

$$1 = \int_{S^3} |\Psi_0|^2 dV \Rightarrow |C|^2 = \frac{1}{2\pi^2}. \quad (29)$$

The wording “constant scalar Laplacian mode” is deliberate. Equation (29) does not describe a spin-1/2 mode or a Dirac zero mode on the positively curved round S^3 .

More importantly, the density in Eq. (29) is convention-dependent. If instead one uses normalized Haar measure

$$d\mu \equiv \frac{dV}{\text{Vol}(S^3)}, \quad \int_{S^3} d\mu = 1, \quad (30)$$

the same constant normalized scalar is simply $\Psi_0 = 1$. Likewise, for a round three-sphere of metric radius ρ ,

$$\text{Vol}(S_\rho^3) = 2\pi^2 \rho^3, \quad (31)$$

and the density relative to the unnormalized Riemannian volume becomes

$$|C_\rho|^2 = \frac{1}{2\pi^2 \rho^3}. \quad (32)$$

Therefore $1/(2\pi^2)$ is not, by normalization alone, a physical coupling-suppression factor. A future dynamics must specify the metric scale, measure, field representation, and canonical field normalization before a volume factor can acquire invariant physical meaning.

7 Postulate E1: electromagnetic correspondence

We now state the central new physical assumption without attributing it to probability normalization:

$$\boxed{\alpha_{\text{TEF}} \equiv \frac{\delta_h}{2\pi^2} = \frac{\sqrt{1+q^2}-1}{2\pi^2}}. \quad (33)$$

Equation (33) is *Postulate E1*. The divisor $2\pi^2$ is the unit-round S^3 volume under Convention C1, but the physical use of that divisor in an electromagnetic coupling is not derived by Convention C1. This distinction is the central limitation of the present paper.

The map can be written directly in terms of the inherited helical angle,

$$\boxed{\alpha_{\text{TEF}} = \frac{\sec \beta - 1}{2\pi^2}}. \quad (34)$$

It can also be written as an explicit gravity-ancestry expression before setting $R = R_G$:

$$\boxed{\alpha_{\text{TEF}}(G, h, c; R) = \frac{\sqrt{1+q(G, h, c; R)^2}-1}{2\pi^2}}, \quad (35)$$

where $q(G, h, c; R)$ is given by Eq. (12). This makes the conditional input chain explicit,

$$G_{\text{exp}} \longrightarrow R_G \longrightarrow q \longrightarrow \alpha_{\text{TEF}}. \quad (36)$$

However, Eq. (35) should not be described as an independent prediction of α from G : once the same gravity calibration $R = R_G$ is imposed, the dimensional constants cancel from q . The final calibrated correspondence becomes

$$\boxed{\alpha_{\text{TEF}} = \frac{1}{2\pi^2} \left[\sqrt{\frac{1 + \sqrt{1 + 16/\pi^2}}{2}} - 1 \right]}, \quad (37)$$

or, more transparently, simply $\alpha_{\text{TEF}} = F(q)$ through

Eq. (33).

8 Discovery-history and mapping-choice burden

Freezing q before the present calculation removes one important form of fitting, but it does not remove functional look-elsewhere freedom. From a single q one can construct many simple quantities, including q , q^2 , $\sin^2 \beta$, δ_h , and combinations involving π or group-space measures. The particular map in Eq. (33) was recognized after comparison with the electromagnetic target. It is therefore not a preregistered or statistically independent prediction even though q itself was inherited without electromagnetic reoptimization.

The present paper freezes Eq. (33) as the object to be tested theoretically. No multiplicative correction factor is introduced, and no alternative function of q is selected to remove the residual in Sec. 10. The meaningful next question is not whether a more accurate expression can be guessed, but whether a TEF action can produce this specific map without post-hoc freedom.

9 Conditional consequence: a correlated electroweak tuple

At tree level the electroweak couplings obey

$$e = g \sin \theta_W = g' \cos \theta_W, \quad \tan \theta_W = \frac{g'}{g}. \quad (38)$$

For the following algebraic comparison we use natural units and Heaviside–Lorentz normalization, $\hbar = c = \varepsilon_0 = 1$, so e denotes a dimensionless gauge coupling rather than the SI charge appearing in the opening definition of α .

Conditional on both candidate identifications $\theta_W \rightarrow \beta$ and $\alpha \rightarrow \alpha_{\text{TEF}}$, define

$$e_{\text{geom}} \equiv \sqrt{4\pi\alpha_{\text{TEF}}} = 0.3031251329\dots, \quad (39)$$

$$g_{\text{geom}} \equiv \frac{e_{\text{geom}}}{\sin \beta} = 0.6235139813\dots, \quad (40)$$

$$g'_{\text{geom}} \equiv \frac{e_{\text{geom}}}{\cos \beta} = 0.3468759044\dots \quad (41)$$

These are not three additional independent predictions. They are algebraic descendants of the two conditional inputs $(\alpha_{\text{TEF}}, \beta)$. Their value is instead one of consistency: any future TEF matching or running mechanism must transform the correlated set jointly rather than repair the fine-structure and weak-angle correspondences independently.

For orientation, current precision electroweak analyses quote scheme-dependent quantities near the Z scale such

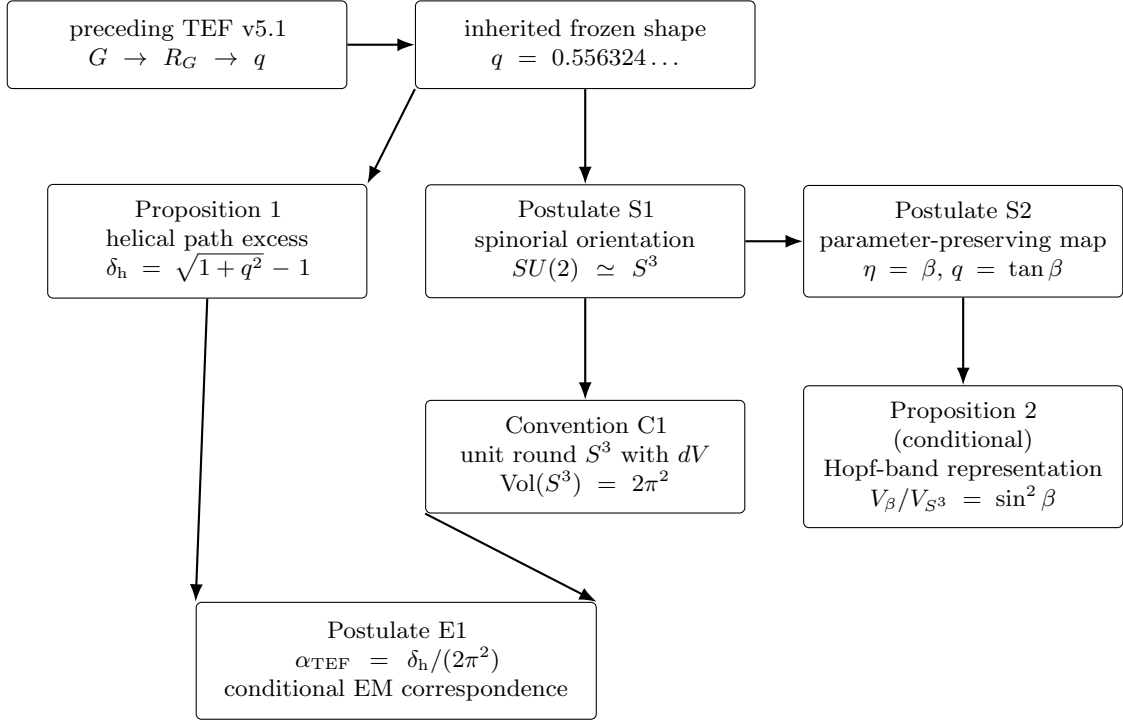


Figure 2: Logical hierarchy of the proposed correspondence. The upper chain contains only the previously released TEF input and its inherited helical shape. The left branch gives the exact helical path excess. The central/right branch separates the spinorial completion, the structural Hopf-coordinate identification, the metric/measure convention, and the conditional weak-angle representation. Postulate E1 receives two distinct inputs—the path excess and the chosen unit-round volume factor—and is therefore displayed separately from Proposition 2. Straight arrows denote inheritance, mathematical implication, or an explicitly named modeling choice; they do not imply that the electromagnetic correspondence has been derived dynamically.

Table 1: Quantities generated after the stated conditional identifications. The last three rows are algebraically correlated descendants, not independent predictions.

Quantity	Geometric value
q	0.556324180045
β	29.0882455°
$\sin^2 \beta$	0.2363477652
δ_h	0.144332378858
α_{TEF}	0.0073119637319
α_{TEF}^{-1}	136.7621663
e_{geom}	0.3031251329
g_{geom}	0.6235139813
g'_{geom}	0.3468759044

as [4]

$$\hat{\alpha}^{(5)}(M_Z^2)^{-1} \simeq 127.93, \quad \sin^2 \hat{\theta}_W(M_Z) \simeq 0.23122. \quad (42)$$

These are not direct targets for Eq. (33); they illustrate why a genuine boundary-condition interpretation requires a unique scale, scheme, threshold prescription, and beta functions.

10 Empirical comparison

Postulate E1 gives

$$\alpha_{\text{TEF}} = 0.007311963731910\dots, \quad (43)$$

$$\alpha_{\text{TEF}}^{-1} = 136.76216631599\dots \quad (44)$$

Relative to the 2022 CODATA recommended value in Eq. (2),

$$\frac{\alpha_{\text{TEF}} - \alpha(0)}{\alpha(0)} = 2.00225595 \times 10^{-3}, \quad (45)$$

or approximately 0.2002256%. This is vastly larger than the uncertainty in the CODATA recommended value and the spread relevant to current precision determinations. Therefore Eq. (33) is not a precision prediction of the measured Thomson-limit coupling.

The scale of the discrepancy can also be compared with direct atom-recoil determinations. Parker *et al.* obtained $\alpha^{-1} = 137.035999046(27)$ from cesium recoil interferometry [5], whereas Morel *et al.* obtained $\alpha^{-1} = 137.035999206(11)$ from rubidium recoil [6]. The two determinations showed a statistically significant tension at their quoted precision, yet their absolute spread is still negligible compared with the 0.2002% TEF residual. That metrological tension therefore cannot be invoked as an explanation of Eq. (45).

The residual is left unchanged. We do not introduce an empirical multiplier, alter q , change the S^3 radius to force agreement, or choose a running scale after inspection of the data. Any future claim that Eq. (33) is a boundary value rather than a low-energy value must supply the matching calculation independently.

11 Relation to current approaches

The idea that gauge couplings might be calculable from ultraviolet physics is not unique to TEF. Eichhorn, Held, and Wetterich showed that asymptotically safe quantum-gravity effects can, for suitable matter content, generate predictive gauge-coupling behavior [10]. Giacometti *et al.* have recently revisited gravitational contributions to gauge and Yukawa beta functions within a proper-time functional flow, explicitly studying gauge-fixing and regulator dependence [11]. Recent work has also explored emergent gauge symmetry and pre-geometric routes to spacetime and gravity [12–14]. These programs differ substantially from TEF in dynamics and mathematical structure; they provide context for the type of ultraviolet question, not support for the present ansatz.

At the opposite, infrared end of the problem, first-principles lattice-QCD calculations now track the hadronic running of both the electromagnetic coupling and the weak mixing angle in the same set of vacuum-polarization observables [7, 8]. This is especially relevant to the correlated TEF tuple in Sec. 9: mainstream precision theory treats the two quantities as jointly scale-dependent electroweak observables. A viable TEF boundary-condition interpretation would therefore need to match and run them together, rather than using separate phenomenological corrections.

Closer to electroweak numerical structure, Meinert and Hofmann obtain approximate electroweak parameters from mixed $SU(2)$ Yang–Mills thermodynamics [16]; Monjo has studied the weak mixing angle within a $U(1, 3)$ colored-gravity construction [17]; Ciborowski investigates geometric constructibility patterns involving electroweak couplings [18]; and Singh places fine-structure and weak-sector normalization in a common octonionic gauge-coupling ledger [19]. Their ontologies and mechanisms are distinct. The present paper claims only a specific conditional map from one earlier gravity-calibrated helical shape to an electromagnetic numerical correspondence.

12 Open dynamical problem and failure conditions

The correspondence in Eq. (33) becomes a physical theory only if a future TEF dynamics removes the conventions and mapping choices exposed above. A schematic target

has the form

$$S_{\text{TEF}} = \int d^4x d\mu(U) \left[\mathcal{L}_{\text{frame}}(q, U) - \frac{1}{4g_*^2} \mathcal{K}_{ab}(q, U) F_{\mu\nu}^a F^{b\mu\nu} + \dots \right]. \quad (46)$$

Equation (46) is not a proposed completed action; it states what the missing calculation must eventually accomplish. In particular, the theory would need to determine: (i) whether the relevant framed configuration space is indeed $SU(2)$; (ii) the metric and physical scale on that space; (iii) the invariant measure and field representations; (iv) canonical mode normalization; (v) the coefficient of the four-dimensional gauge kinetic term; (vi) the relation between the electromagnetic and weak gauge sectors; and (vii) a common renormalization-group and threshold-matching map.

The present correspondence has explicit theoretical failure conditions. It fails as an explanation of α if no consistent TEF action produces Postulate E1, if the physically selected metric or measure removes the proposed factor, if the $\eta = \beta$ identification cannot be derived from the framed geometry, or if a unique joint electroweak matching map conflicts with experiment. Quantitative experimental falsifiability as an electroweak boundary condition requires those missing scale, scheme, and RG specifications first.

13 Logical status of the result

The hierarchy of the present paper is intentionally explicit:

Inherited result	gravity-calibrated q from Ref. [1]
Exact geometry	$\delta_h = \sqrt{1 + q^2} - 1$
Postulate S1	framed $SU(2) \simeq S^3$ completion
Postulate S2	parameter-preserving $\eta = \beta$ map
Exact consequence	$V_\beta/V_{S^3} = \sin^2 \beta$
Convention C1	unit round metric, unnormalized dV
Postulate E1	$\alpha_{\text{TEF}} = \delta_h/(2\pi^2)$
Conditional consequence	correlated electroweak tuple

This separation is important methodologically. The paper does not derive the fine-structure constant. It identifies a compact numerical correspondence whose upstream geometric parameter is already fixed by an earlier gravity-calibrated ansatz, and it states precisely which additional assumptions are needed to turn that correspondence into physics.

14 Conclusion

The preceding TEF construction fixes a dimensionless helical shape through the chain

$$G_{\text{exp}} \rightarrow R_G \rightarrow q, \quad q\sqrt{1+q^2} = \frac{2}{\pi}, \quad (47)$$

without electroweak input. The present paper shows that the same q generates the scale-free helical path excess

$$\delta_h = \sqrt{1+q^2} - 1 \quad (48)$$

and, conditional on a spinorial $SU(2) \simeq S^3$ completion together with the minimal identification $\eta = \beta$, gives the exact Hopf-band representation

$$\frac{V_\beta}{V_{S^3}} = \sin^2 \beta. \quad (49)$$

Under the additional and explicitly named electromagnetic Postulate E1,

$$\boxed{\alpha_{\text{TEF}} = \frac{\sqrt{1+q^2} - 1}{2\pi^2}}, \quad (50)$$

which yields $\alpha_{\text{TEF}}^{-1} = 136.7621663 \dots$. The result lies about

0.2002% from the 2022 CODATA recommended low-energy value. No correction is applied.

The main conclusion is therefore not that the fine-structure constant has been derived. It is that an earlier, publicly frozen, gravity-calibrated helical parameter admits a particularly compact conditional map to the electromagnetic numerical regime. Before the gravity calibration is imposed the map may be written as $\alpha_{\text{TEF}}(G, h, c; R)$ through Eq. (35); after calibration, the dimensional constants cancel and the same proposal is most cleanly expressed as $\alpha_{\text{TEF}} = F(q)$. Whether this is a physically meaningful bridge from gravity-calibrated microscopic geometry to electroweak coupling, or only a structured numerical coincidence, is now reduced to a concrete dynamical question: can the orientation space, metric, measure, gauge kinetic normalization, and joint electroweak matching be derived from one TEF action?

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